



Branching as Active Inheritance

A Coherence-Threshold Account of History-Bearing Organization Across Morphology



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Branching and ramified organization recur across river basins, roots, fungal mycelia, vascular systems, epithelial organs, adaptive transport networks, planetary drainage landscapes, and three-dimensional fractal aggregates. In each, earlier flow, growth, or transport can alter physical structure, and the retained structure can continue shaping later dynamics. Branching as Active Inheritance treats that relation as a general principle of history-bearing organization, treats its maintenance as a rate-balance problem measured by the Porter Ratio, and states how to test whether R predicts what branching systems keep and reuse.

What branching systems share

What do branching systems as different as rivers, roots, and blood vessels share, and can one measured quantity predict how they carry their own history forward?

How branching carries history

1. **Earlier dynamics change structure, and retained structure shapes later dynamics.** Across branching systems, earlier flow, growth, or transport can alter physical structure, and the retained structure can continue shaping later dynamics.
2. **Three linked concepts.** Active inheritance names the continued causal participation of retained organization. Consequential geometry names the spatial structure through which earlier dynamics constrain later dynamics. Consequential history names the accumulation of those retained constraints through time.
3. **Maintenance is a rate-balance problem.** Maintenance of inherited organization is a balance between two rates, λ_{self} and λ_{env} , compared by the Porter Ratio.
4. **What R predicts.** Independently measured R predicts structural recovery, path persistence, route reuse, topology retention, and dependence on prior state, and the test is built to check exactly this.
5. **Organizational invariance across morphology.** Dimensionality, boundary conditions, gravity, pressure, material properties, gradients, and forcing shape the visible form through which organization is expressed. The invariant is the causal relation by which earlier dynamics alter structure and retained structure influences what happens next. Its visible morphology can change while the history-bearing relation remains measurable.
6. **Beyond branching.** Active inheritance extends into organizational ancestry and into interiority. Phenomenal consciousness belongs to a later stage: sufficiently deep and integrated recursive availability of inherited internal state.

Terms to know

- **Active inheritance.** Active inheritance is the continued causal participation of earlier organization within later organization. The carrier can change while the consequence continues. Here: the continued causal participation of retained organization.
- **Consequential geometry.** The spatial structure through which earlier dynamics constrain later dynamics. In plain terms: the shape left behind by past flow, which

steers future flow.

- **Consequential history.** Consequential history is the portion of the past that remains causally active in the present. Here: the accumulation of retained constraints through time.
- **Organizational ancestry.** Causal continuity in retained organization across physical and biological history.
- **Interiority.** Graded causal self-conditioning as inherited organization becomes increasingly local, nested, and recursively available to present dynamics. In plain terms: the degree to which a system’s own carried organization shapes what happens to it.

The numbers

$R = \lambda_{\text{self}} / \lambda_{\text{env}}$

- λ_{self} is the effective rate at which a selected organization is restored, reinforced, or reliably propagated.
- λ_{env} is the effective rate at which current forcing revises that same organization.

Both rates concern the same organizational variable over the same interval, so R is dimensionless.

- **R = 1** marks equality of the two rates.
- **Sustained R > 1** defines maintenance dominance for the selected organizational variable and interval.

Where it shows up

- **Rivers, roots, fungi, blood vessels.** Earlier flow or growth carves or builds a structure, and that structure channels later flow or growth.
- **Surface and volume.** Surface-constrained transport can produce dendritic branching, while three-dimensional growth can produce volumetric ramification with a distinct radial, approximately spherical, or ellipsoidal envelope.
- **Mars and Titan.** Martian and Titanian drainage networks show dendritic organization under planetary conditions substantially different from those of Earth.
- **Three-dimensional aggregates.** Three-dimensional diffusion-limited aggregation shows that internal ramification and overall envelope can be described separately.

How to test it

A rigorous test:

1. defines the organizational variable and both rates in advance;
2. measures λ_{self} and λ_{env} independently;
3. calculates R before the scored outcome;
4. compares the predictive value of R with geometry, λ_{self} alone, λ_{env} alone, their difference, and established domain-specific predictors.

Replication across different morphologies tests whether the same maintenance relation holds through changes in visible form.

What experiments can check

Independently measured R predicts:

- structural recovery;
- path persistence, whether channels keep their routes;

- route reuse;
- topology retention, whether the branching pattern keeps its connections;
- dependence on prior state.

The bigger picture

The same causal structure runs through rivers, roots, fungi, and blood vessels. Earlier flow or growth builds structure, retained structure guides later flow or growth, and the Porter Ratio measures how well that inherited organization is kept.

Full paper on PhilArchive: [Branching as Active Inheritance: *A Coherence-Threshold Account of History-Bearing Organization Across Morphology*](#)

Before this: [Aleph Harmonic Qualia: The Dynamical Click of Coherence](#).

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